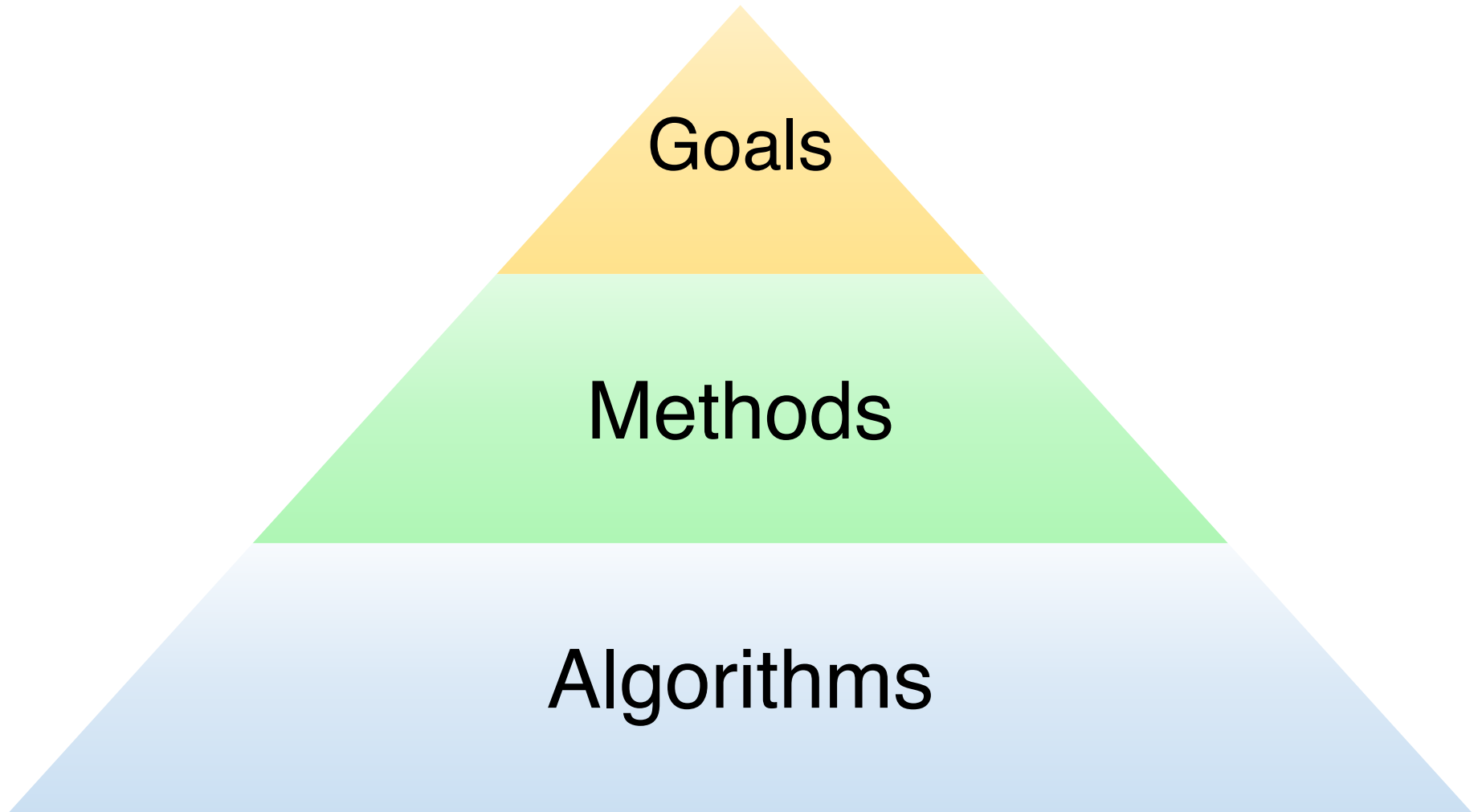


Hierarchy of Data Mining



Goals of Data Mining

“The knowledge discovery *goals* are defined by the intended use of the system. We can distinguish two types of goals: **Verification**, where the system is limited to verifying the user’s hypothesis, and **Discovery**, where the system autonomously finds new patterns.”

Goals of Data Mining

“We further subdivide the **Discovery** goal into **Prediction**, where the system finds patterns for the purpose of predicting the future behaviour of some entities; and **Description**, where the system finds patterns for the purpose of presenting them to a user in a human-understandable form.”

Basic Data Mining Methods

- Classification
- Estimation
- Affinity Grouping
- Clustering
- Description
- Prediction

Classification

- Examining the features of an object and assigning it to one of a predefined set of classes
- Eg. Classifying credit applicants as low, medium, or high risk – HOW?
- Algorithms: decision trees, memory-based reasoning, link analysis
- Discrete outcomes – YES/NO, LOW/HIGH, MEASLES/RUBELLA/CHICKEN POX,...

Estimation

- “Given some input data we use estimation to come up with a value for some unknown continuous variable such as income, height, or credit balance”
- Algorithms: Neural Networks

Affinity Grouping/Market Basket Analysis

- “to determine which things go together” so that “items purchased together will be seen together”, “cross-selling”, “attractive packages”, “groups of product and services”
- Eg. “if a customer purchases a razor and after shave, then that customer will purchase shaving cream with 80% confidence” - **if** *antecedent(s)* **then** *consequent(s)*
- Algorithms: Association Rules – GRI, Apriori

Clustering

- “segmenting a heterogeneous population into a number of more homogeneous subgroups or clusters”
- “no predefined classes and no examples”
- Undirected knowledge discovery (KD) or unsupervised learning
- Results of undirected KD (clusters) become the input for more directed KD
- Algorithms: Kohonen Networks or self-organizing map, K-means, Two Step Cluster

Description

- Description of the data
- “a good enough description of a behavior will often suggest an explanation for it as well”
- Eg. Average age, percent female, people who see movie X also see movie Y, males aged 12 to 17 are likely to see movie X.
- Algorithms: standard summary statistics, affinity grouping, rules.

Prediction

- “Same as classification or estimation except that the records are classified according to some predicted future behavior or estimated future value”
- Algorithms: Decision Trees, Neural Networks, Regression analysis, Correlation Analysis, Market Basket Analysis, Memory-based Reasoning
- Eg. Predicting which customers will leave in the future

Other Points to consider

- Training, Test, & Evaluation Datasets
- Importance of outliers in data mining
- How much data? At least 10,000 pre-classified records.
- How many variable? As many as possible.
- Problem of Overfitting